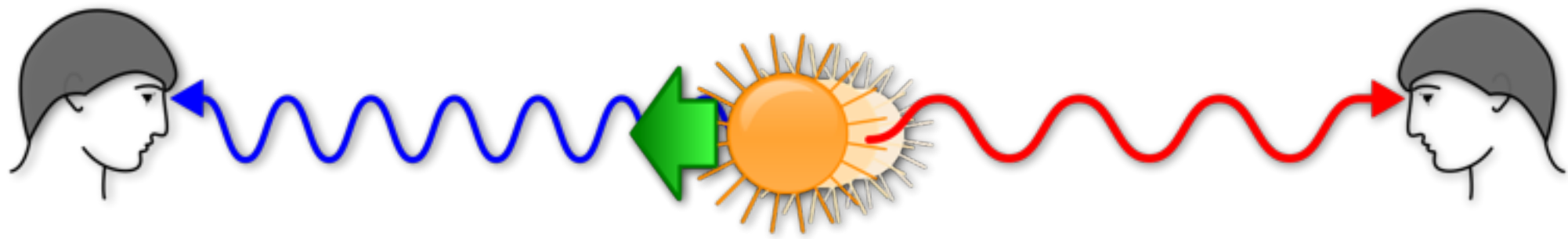


Redshift



Slow and Big

FRW Metric

- **The universe is homogeneous and isotropic:** As the universe is filled with matter and radiation, comoving observers, who are at rest with respect to this matter and radiation, perceive the universe as such.
- **The universe is homogeneous and isotropic *at all times*:** The metric describing the expanding space is invariant under transformations that preserve distances and angles (isometries). There are only three possible spatial geometries that satisfy homogeneity and isotropy:
 - Flat space (Euclidean, zero curvature).
 - Three-dimensional sphere (constant positive curvature),
 - Three-dimensional hyperboloid (constant negative curvature).
- Consider, for example, a sphere (two-sphere) in 3D space $x^2 + y^2 + z^2 = R^2$.

Since, $dz = -\frac{xdx + ydy}{z} = -\frac{xdx + ydy}{\sqrt{R^2 - x^2 - y^2}}$, we have

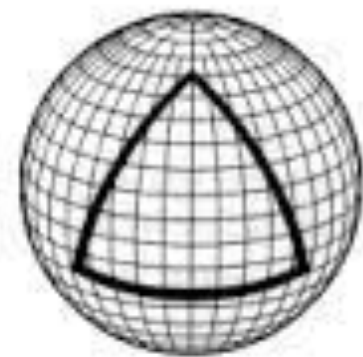
$$ds^2 = (dx)^2 + (dy)^2 + \frac{(xdx + ydy)^2}{R^2 - x^2 - y^2}$$

FRW Metric

- Let $x = r' \cos \theta$, $y = r' \sin \theta$. This gives,

$$ds^2 = \frac{dr'^2}{1 - r'^2/R^2} + r'^2 d\theta^2$$

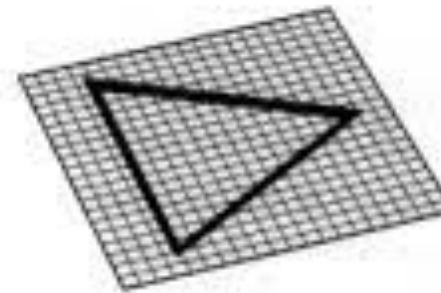
- $0 < R^2 < \infty$ corresponds to the on a sphere; $-\infty < R^2 < 0$ to a hyperbolic surface; $R^2 \rightarrow \infty$ to a flat surface.



Positive Curvature



Negative Curvature



Flat Curvature

- We can now extend this to a three-surface embedded in 4D. Introducing a fourth orthogonal direction ω , the metric is

$$ds^2 = (dx)^2 + (dy)^2 + (dz)^2 + (d\omega)^2$$

FRW Metric

- Consider a homogeneous and isotropic three-hypersurface of radius R :

$$R^2 = x^2 + y^2 + z^2 + \omega^2; \quad d\omega = -\frac{xdx + ydy + zdz}{\sqrt{R^2 - x^2 - y^2 - z^2}}$$

- Choosing (r', θ, ϕ) such that

$$x = r' \sin \theta \cos \phi, \quad y = r' \sin \theta \sin \phi, \quad z = r' \cos \theta$$

- We get

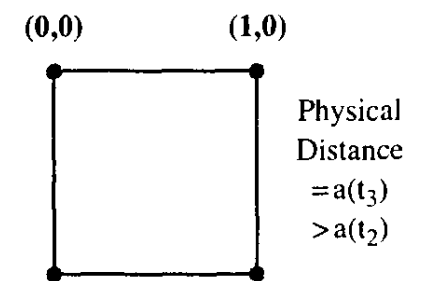
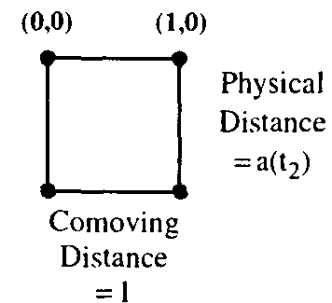
$$ds^2 = \frac{dr'^2}{1 - r'^2/R^2} + r'^2(d\theta^2 + \sin^2 \theta d\phi^2)$$
$$ds^2 = R^2 \left(\frac{dr^2}{1 - Kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right) \text{ where } K \text{ is } \pm 1 \text{ or } 0.$$

That is: $g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a(t)^2}{1 - Kr^2} & 0 & 0 \\ 0 & 0 & a(t)^2 r^2 & 0 \\ 0 & 0 & 0 & a(t)^2 r^2 \sin^2 \theta \end{pmatrix}$ where $a(t) = R$ (scale factor).

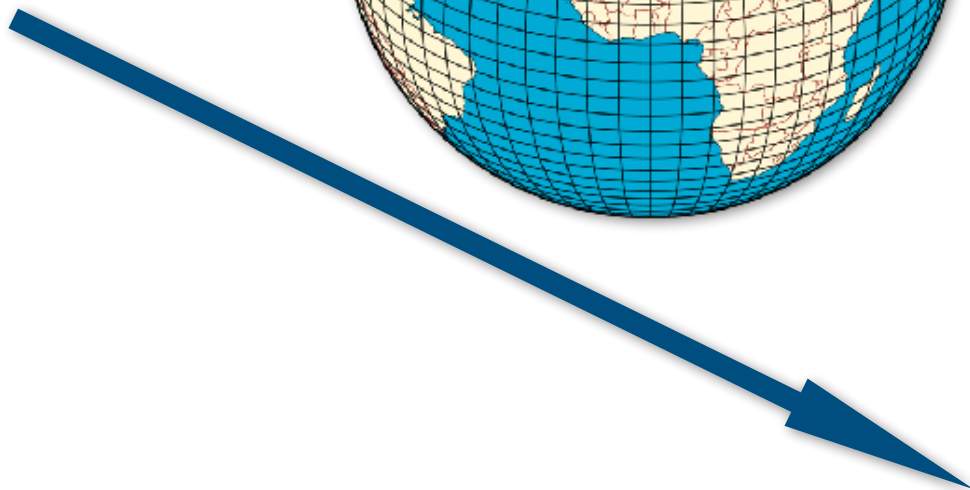
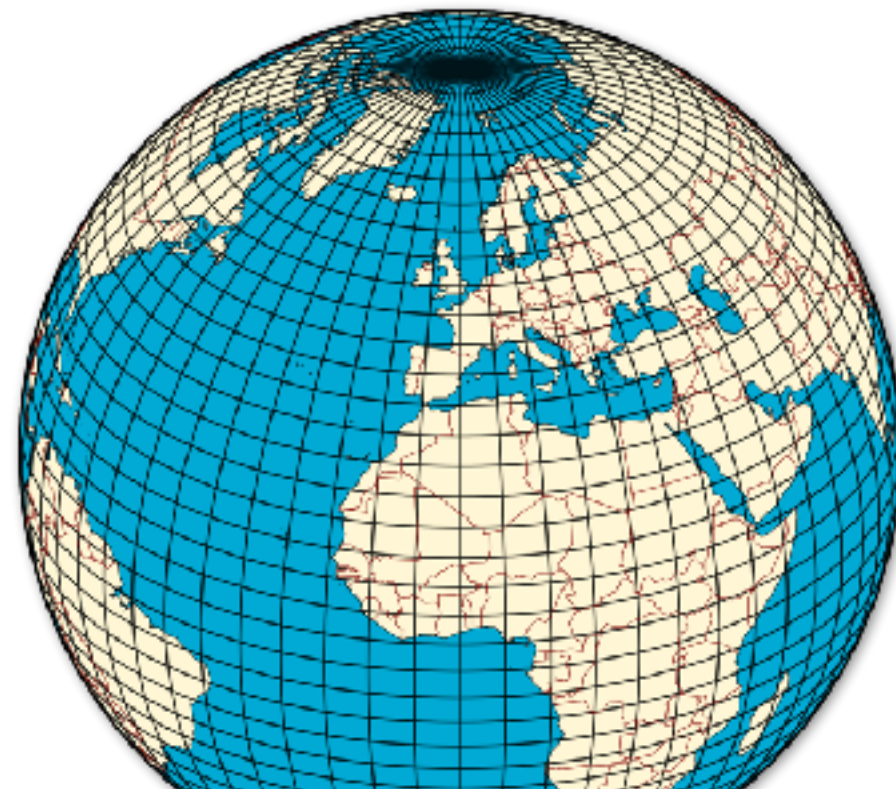
FRW Metric

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a(t)^2}{1 - Kr^2} & 0 & 0 \\ 0 & 0 & a(t)^2 r^2 & 0 \\ 0 & 0 & 0 & a(t)^2 r^2 \sin^2 \theta \end{pmatrix}$$

where $a(t) = R$ (scale factor).



Time →



Cosmological Redshift

- Suppose an observer at $(0,0,0)$ observes light from the galaxy at $(r,0,0)$. A light wave travels along the null geodesic $ds^2 = 0$,

$$dt^2 = a(t)^2 \frac{dr^2}{1 - Kr^2} \implies dt = -a(t) \frac{dr}{\sqrt{1 - Kr^2}}$$

we take the negative sign as the distance from the observer decreases with time.

- We get

$$\int_{t_e}^{t_o} \frac{dt}{a(t)} = - \int_r^0 \frac{dr}{\sqrt{1 - Kr^2}} = \int_0^r \frac{dr}{\sqrt{1 - Kr^2}} = C \text{ (a constant)}$$

- Now consider two waves consecutively emitted by the galaxy at times t_e and $t_e + \delta t_e$ and observed at times t_o and $t_o + \delta t_o$:

$$\int_{t_e}^{t_o} \frac{dt}{a(t)} = \int_{t_e + \delta t_e}^{t_o + \delta t_o} \frac{dt}{a(t)} = C$$

Cosmological Redshift

- Subtracting $\int_{t_e+\delta t_e}^{t_o} dt/a(t)$ from both sides gives:

$$\int_{t_e}^{t_e+\delta t_e} \frac{dt}{a(t)} = \int_{t_o}^{t_o+\delta t_o} \frac{dt}{a(t)}$$

- For sufficiently small intervals δt_e and δt_o , the scale factor $a(t)$ remains roughly constant: $a(t_e) = a(t_e + \delta t_e)$ and $a(t_o) = a(t_o + \delta t_o)$.
- So we get

$$\frac{\delta t_e}{a(t_e)} = \frac{\delta t_o}{a(t_o)}$$

- δt_e is the time interval between two consecutive crests being emitted by the distant galaxy and δt_o is the time interval between two consecutive crests being observed. If the emitted and detected waves have frequency and wavelengths as ν_e, λ_e and ν_o, λ_o , respectively, then $\nu_e = 1/\delta t_e$ and $\nu_o = 1/\delta t_o$.

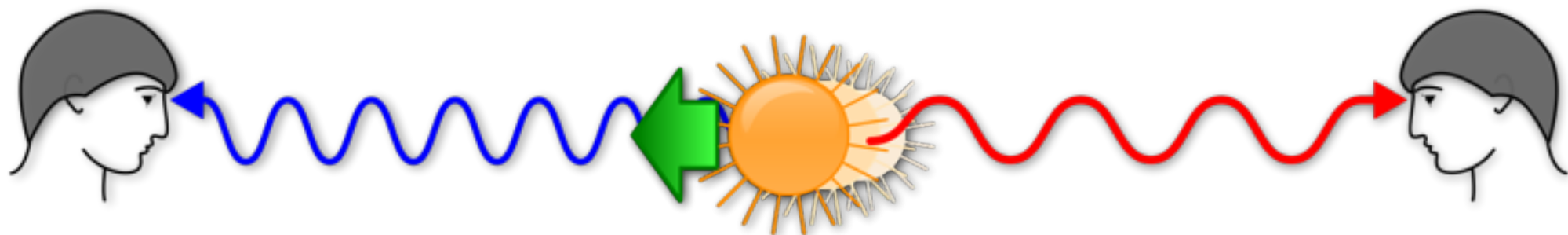
Cosmological Redshift

- This gives $\frac{\nu_o}{\nu_e} = \frac{a(t_e)}{a(t_o)} \implies \frac{\lambda_e}{\lambda_o} = \frac{a(t_e)}{a(t_o)}$
- The redshift of the galaxy is defined as

$$z = \frac{\lambda_o - \lambda_e}{\lambda_e} = \frac{\lambda_o}{\lambda_e} - 1 = \frac{a(t_o)}{a(t_e)} - 1$$

$$\text{or } 1 + z = \frac{a(t_o)}{a(t_e)}$$

- If $z > 0$, known as redshift, $a(t_o) > a(t_e)$ which indicates expansion. If $z < 0$, called blueshift, $a(t_o) < a(t_e)$ which indicates contraction.



Expansion Dynamics

- The dynamical evolution of the universe is captured by the **Friedmann's equations** follow from the **Einstein's field equation**:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G'T_{\mu\nu}.$$

Einstein later added an extra constant term to describe a static universe

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G'T_{\mu\nu}$$

- After Hubble's discovery of the expansion of the universe Einstein regretted adding this Λ term calling it his greatest blunder. But as we now know, the universe does have a cosmological constant. It is interpreted as the **energy density of the vacuum**.
- Quantum theory predicts that $\Lambda \sim m_{\text{PL}}^4$ where $m_{\text{PL}} = 10^{19}$ GeV is the Planck mass. But experimental observation of Λ is at least a factor of 10^{120} smaller. This is the largest known discrepancy in physics, and this problem is called the **cosmological constant problem**. Intuitively, the value of Λ is expected to be small since in the non-relativistic limit the field equation must give the Newton's 2nd law of gravitation.

Friedmann equations

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{K}{a^2} \quad \text{and} \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) + \frac{\Lambda}{3}$$

- The second equation, known as the acceleration equation, describes the rate of change of the expansion. Note that a positive cosmological constant Λ **accelerates** the expansion of the universe.
- To fully determine the dynamics, one must also know how mass and energy density evolve over time. The evolution of energy density is governed by the fluid equation.

$$\frac{\partial \rho}{\partial t} + \frac{\dot{a}}{a}(3\rho + 3P) = 0$$

- By defining the equation of state parameter $\omega = P/\rho$, the fluid equation yields a solution:

$$\rho \propto a^{-3(1+\omega)} \quad (\text{the equation of state})$$

Different matters

The three types of matter frequently encountered in the universe are:

- **Cold matter/dust/non-relativistic matter:** Non-relativistic particles have negligible momenta, so for non-relativistic matter, the pressure $P = 0$, and $\omega = 0$. Thus, the equation of state takes the form: $\rho \propto a^{-3}$.
- **Radiation/hot matter:** Radiation is characterised by masslessness. In cosmology, radiation includes photons, neutrinos etc. The pressure for radiation is given by $P = \rho/3$, so that the equation of state becomes: $\rho \propto a^{-4}$.
- **Cosmological constant:** By taking the cosmological constant term to the right hand side, it can be interpreted as the energy associated with the vacuum. This type of matter/energy is unusual because it has negative pressure:

$$-P = \rho_{\Lambda} = \frac{\Lambda}{8\pi G}$$

Critical density

- Absorbing the ρ_Λ into ρ , the Friedmann equations can be written as

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{K}{a^2} \quad \text{and} \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P)$$

- Critical density and K:

$$1 = \frac{\rho}{3H^2/8\pi G} - \frac{K}{a^2H^2} = \frac{\rho}{\rho_{crit}} - \frac{K}{a^2H^2}$$

For any value of H , i.e., at any epoch of the cosmic evolution, there is a critical

density of the universe: $\rho_{crit} = \frac{3H^2}{8\pi G}$

- $\rho = \rho_{crit}$ then $K = 0$; the universe is flat.
- $\rho > \rho_{crit}$ then $K > 0$; the universe is closed.
- $\rho < \rho_{crit}$ then $K < 0$; the universe is open

Critical density

- At the present epoch t_0 , the Hubble constant is $H_0 = 72$ km/s/Mpc and the corresponding critical density is: $\rho_{crit,0} = \frac{3H_0^2}{8\pi G} = 1.9h^2 \times 10^{-29} \text{ g cm}^{-3}$ where $h = 0.72$.

- There is something strange associated with the critical density of the universe. The energy density ρ of mass and radiation scales as $1/a^3$ and $1/a^4$, respectively, while ρ_Λ remains constant. If we extrapolate to early times, ρ would be large and dominated by radiation, making the $-K/a^2$ term negligible. This implies:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho \Rightarrow \left[1 = \frac{\rho}{\rho_{crit,0}}\right]$$

- It means the density of the universe was equal to its critical density at very early times. Experimental observations show that even today, $\rho_0 \approx \rho_{0,crit}$, implying a flat $K = 0$ universe. This is a puzzle because billion years of evolution should have made ρ_0 significantly different from $\rho_{crit,0}$; this is referred to as the flatness problem.